Elementary Probability

Moments, Conditioning, Bayes, and Generating Functions

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 1

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1 Mean and Variance

1.1 Definitions

For a discrete random variable X with pmf $p(x) = P(X = x)$, and for a continuous X with density $f(x)$, the **expected value** is

$$\mathbb{E}[X] = \mu = \sum_x x\, p(x) \qquad \text{or} \qquad \mathbb{E}[X] = \mu = \int_{-\infty}^{\infty} x f(x)\, dx,$$

provided the sum or integral converges absolutely. The **variance** is

$$\operatorname{Var}(X) = \sigma^2 = \mathbb{E}\big[(X-\mu)^2\big] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2\,,$$

with standard deviation $\sigma = \sqrt{\operatorname{Var}(X)}$.

1.2 Reference Distributions

Distribution	pmf / pdf	Mean	Variance
Binomial(n,p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$
Poisson(λ)	$\lambda^k e^{-\lambda}/k!$	λ	λ
Normal(μ,σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2
Exponential(λ)	$\lambda e^{-\lambda x},\ x\geq 0$	$1/\lambda$	$1/\lambda^2$

The Poisson is the one common case where mean and variance coincide; departures from $\mathrm{Var}/\mathbb{E}=1$ in count data (overdispersion) are the usual reason to move to a negative binomial.

Derivation (exponential mean). Integrating by parts with $u=x$, $dv=\lambda e^{-\lambda x}dx$,

$$\mathbb{E}[X]=\int_0^\infty x\lambda e^{-\lambda x}\,dx=\Big[-xe^{-\lambda x}\Big]_0^\infty+\int_0^\infty e^{-\lambda x}dx=\frac{1}{\lambda}.$$

2 Expectation and Variance Theory

2.1 Linearity

Linearity of expectation. For any random variables $X_1,\dots,X_n$ and constants $a_1,\dots,a_n,b$,

$$\mathbb{E}\Big[\sum_{i=1}^n a_iX_i+b\Big]=\sum_{i=1}^n a_i\mathbb{E}[X_i]+b.$$

This holds *regardless of dependence*, which is what makes expectation far easier to work with than variance.

2.2 Nonnegativity of Variance

For any X with a finite second moment, $\mathbb{E}[X^2]\geq(\mathbb{E}[X])^2$, with equality if and only if X is almost surely constant. This is Jensen's inequality applied to $x\mapsto x^2$, and it is equivalent to the statement $\mathrm{Var}(X)\geq 0$:

$$0\leq\mathbb{E}\big[(X-\mathbb{E}[X])^2\big]=\mathbb{E}[X^2]-2(\mathbb{E}[X])^2+(\mathbb{E}[X])^2=\mathbb{E}[X^2]-(\mathbb{E}[X])^2.$$

2.3 Moment Hierarchy

Moment hierarchy. If $\mathbb{E}[|X|^k]<\infty$ for a positive integer k, then $\mathbb{E}[|X|^j]<\infty$ for every positive integer $j\leq k$.

Proof. Split on $A=\{|X|\leq 1\}$ and $B=\{|X|>1\}$. On A, $|X|^j\leq 1$, so $\mathbb{E}[|X|^j\mathbf{1}_A]\leq P(A)\leq 1$. On B, $|X|^j\leq|X|^k$ because $j<k$, so $\mathbb{E}[|X|^j\mathbf{1}_B]\leq\mathbb{E}[|X|^k]<\infty$. Adding the two pieces gives the result. □

The converse fails, and the failure matters in finance. The Cauchy density $f(x)=1/\big(\pi(1+x^2)\big)$ has *no* finite moment of any positive order, so the hierarchy breaks at the first level. The Pareto density $f(x)=\alpha x^{-(\alpha+1)}$, $x\geq 1$, has finite kth moments only for $k<\alpha$ — and estimated tail indices for daily equity returns sit near $\alpha\approx 3$–4, which is precisely the regime where the variance is finite but the kurtosis is not reliably estimable.

2.4 Variance, Covariance, Correlation

$$\mathrm{Var}(aX+b)=a^2\,\mathrm{Var}(X),\qquad \mathrm{Var}(X+Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)+2\,\mathrm{Cov}(X,Y),$$

with the cross term vanishing under independence. The covariance and correlation are

$$\mathrm{Cov}(X,Y)=\mathbb{E}\big[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\big]=\mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y],\qquad \rho(X,Y)=\frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}\in[-1,1].$$

Portfolio application. For weights w_i and asset returns R_i, the portfolio return $R_p=\sum_i w_iR_i$ has

$$\mathrm{Var}(R_p)=\sum_{i=1}^n w_i^2\,\mathrm{Var}(R_i)+2\sum_{i<j}w_iw_j\,\mathrm{Cov}(R_i,R_j).$$

The second term is the whole content of diversification: with n equally weighted assets of common variance σ^2 and common pairwise correlation ρ, the portfolio variance tends to $\rho\sigma^2$ as $n\to\infty$ rather than to zero.

2.5 Conditional Expectation

$$\mathbb{E}[X \mid Y = y] = \sum_x x\, P(X = x \mid Y = y) \qquad \text{or} \qquad \mathbb{E}[X \mid Y = y] = \int_{-\infty}^{\infty} x\, \frac{f_{X,Y}(x,y)}{f_Y(y)}\, dx.$$

Note that $\mathbb{E}[X \mid Y]$ is itself a random variable — a function of Y — which is what makes the two identities of the next section possible.

3 Conditioning Identities

3.1 Law of Total Expectation (Adam's Law)

Tower property. For random variables X and Y,

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big].$$

If $\{B_1, B_2, \dots\}$ partitions the sample space, the discrete form is $\mathbb{E}[X] = \sum_i \mathbb{E}[X \mid B_i]\, P(B_i)$.

Proof (discrete). $\mathbb{E}[X] = \sum_x xP(X = x) = \sum_x x \sum_i P(X = x \mid B_i)P(B_i) = \sum_i P(B_i) \sum_x xP(X = x \mid B_i) = \sum_i P(B_i)\mathbb{E}[X \mid B_i]$. □

Example. An insurer writes high-risk policies (30% of the book, expected claim \$2000) and low-risk policies (70%, expected claim \$500). The expected claim on a randomly selected policy is $\mathbb{E}[X] = 2000(0.3) + 500(0.7) = \$950$.

3.2 Law of Total Variance (Eve's Law)

Variance decomposition.

$$\operatorname{Var}(X) = \underbrace{\mathbb{E}\big[\operatorname{Var}(X \mid Y)\big]}_{\text{average within-group variance}} + \underbrace{\operatorname{Var}\big(\mathbb{E}[X \mid Y]\big)}_{\text{variance of group means}}.$$

Proof. Write $X - \mathbb{E}[X] = \big(X - \mathbb{E}[X \mid Y]\big) + \big(\mathbb{E}[X \mid Y] - \mathbb{E}[X]\big)$ and square. The cross term vanishes by conditioning on Y:

$$\mathbb{E}\big[(X - \mathbb{E}[X \mid Y])(\mathbb{E}[X \mid Y] - \mathbb{E}[X])\big] = \mathbb{E}\Big[(\mathbb{E}[X \mid Y] - \mathbb{E}[X])\, \underbrace{\mathbb{E}\big[X - \mathbb{E}[X \mid Y] \,\big|\, Y\big]}_{=0}\Big] = 0. \qquad \square$$

Example. Machine A makes 60% of parts (mean 10 cm, sd 0.5), machine B makes 40% (mean 12 cm, sd 0.8). Then $\mathbb{E}[\operatorname{Var}(X \mid Y)] = 0.6(0.25) + 0.4(0.64) = 0.406$, and with $\mathbb{E}[X] = 10.8$ and $\mathbb{E}\big[(\mathbb{E}[X \mid Y])^2\big] = 0.6(100) + 0.4(144) = 117.6$ we get $\operatorname{Var}(\mathbb{E}[X \mid Y]) = 117.6 - 10.8^2 = 1.04$. Hence $\operatorname{Var}(X) = 0.406 + 1.04 = 1.446$.

3.3 Iterated Conditioning

For random variables X, Y, Z,

$$\mathbb{E}[X \mid Y] = \mathbb{E}\big[\,\mathbb{E}[X \mid Y, Z] \,\big|\, Y\,\big].$$

Conditioning on the finer information first and then averaging it out recovers the coarser conditional expectation. This is the identity that underlies martingale arguments and every recursive valuation scheme.

4 Bayes' Theorem

For events A and B with $P(B) > 0$, and for a partition $\{B_1, \dots, B_n\}$,

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}, \qquad P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{n} P(A \mid B_j)P(B_j)}.$$

Here $P(B_i)$ is the **prior**, $P(A \mid B_i)$ the **likelihood**, $P(B_i \mid A)$ the **posterior**, and the denominator the **marginal likelihood** that normalises it. Applied sequentially, each posterior becomes the next prior: $P(B \mid A_1, A_2) \propto P(A_2 \mid B, A_1)P(B \mid A_1)$.

Base rates dominate. A disease affects 1% of the population; a test has sensitivity $P(T^+ \mid D) = 0.95$ and specificity $P(T^- \mid D^c) = 0.99$. Then

$$P(D \mid T^+) = \frac{0.95(0.01)}{0.95(0.01) + 0.01(0.99)} = \frac{0.0095}{0.0194} \approx 0.49.$$

A highly accurate test yields a coin flip, because the false positives drawn from the large healthy population are comparable in number to the true positives.

Spam filter. 40% of email is spam, 80% of spam contains "free", 10% of legitimate mail does. Then $P(\text{spam} \mid \text{free}) = 0.32/(0.32 + 0.06) \approx 0.84$.

5 Moment Generating Functions

5.1 Definition and Properties

The **moment generating function** of X is $M_X(t) = \mathbb{E}[e^{tX}]$, wherever the expectation exists — explicitly $\sum_x e^{tx} p(x)$ or $\int e^{tx} f(x)\,dx$.

If M_X exists in a neighbourhood of the origin, then

1. **Moments:** $\mathbb{E}[X^n] = M_X^{(n)}(0)$.
2. **Uniqueness:** M_X determines the distribution of X.
3. **Independence:** $X \perp Y \implies M_{X+Y}(t) = M_X(t) M_Y(t)$.

Proof of (1). Expanding the exponential and exchanging sum and expectation, $M_X(t) = \mathbb{E}\big[\sum_n (tX)^n/n!\big] = \sum_n t^n \mathbb{E}[X^n]/n!$. Differentiating n times and setting $t = 0$ leaves $\mathbb{E}[X^n]$. **Proof of (3).** $\mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX} e^{tY}] = \mathbb{E}[e^{tX}]\mathbb{E}[e^{tY}]$ by independence. □

5.2 Common MGFs

Distribution	MGF $M_X(t)$	Domain
Bernoulli(p)	$1 - p + pe^t$	$t \in \mathbb{R}$
Binomial(n, p)	$(1 - p + pe^t)^n$	$t \in \mathbb{R}$
Poisson(λ)	$e^{\lambda(e^t - 1)}$	$t \in \mathbb{R}$
Geometric(p)	$pe^t / \big(1 - (1-p)e^t\big)$	$t < -\ln(1-p)$
Normal(μ, σ^2)	$e^{\mu t + \sigma^2 t^2/2}$	$t \in \mathbb{R}$
Exponential(λ)	$\lambda/(\lambda - t)$	$t < \lambda$
Gamma(α, λ)	$\big(\lambda/(\lambda - t)\big)^{\alpha}$	$t < \lambda$

For the exponential, $M_X(t) = \lambda \int_0^\infty e^{(t-\lambda)x} dx = \lambda/(\lambda - t)$, convergent only for $t < \lambda$ — the finite domain is the signature of an exponentially decaying tail, and a distribution with a heavier tail has no MGF at all.

5.3 Closure Under Convolution

Uniqueness plus the product rule identifies the distribution of a sum in one line. For independent $X_i \sim N(\mu_i, \sigma_i^2)$,

$$M_{X_1+X_2}(t) = e^{\mu_1 t + \sigma_1^2 t^2/2}\, e^{\mu_2 t + \sigma_2^2 t^2/2} = e^{(\mu_1+\mu_2)t + (\sigma_1^2+\sigma_2^2)t^2/2},$$

so $X_1 + X_2 \sim N(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2)$. Identically, for independent Poissons, $M_{X+Y}(t) = e^{(\lambda_1+\lambda_2)(e^t-1)}$, so $X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$.

6 Characteristic Functions

6.1 Definition

The **characteristic function** of X is

$$\phi_X(t) = \mathbb{E}[e^{itX}] = \mathbb{E}[\cos tX] + i\,\mathbb{E}[\sin tX],$$

the Fourier transform of the distribution. Unlike the MGF, it *always* exists, because $|e^{itX}| = 1$ makes the expectation automatically finite. When the MGF exists in a neighbourhood of the origin the two are related by $\phi_X(t) = M_X(it)$.

Properties. For any random variable X and all real t:

1. $\phi_X(0) = 1$ and $|\phi_X(t)| \le 1$;
2. $\phi_X(-t) = \overline{\phi_X(t)}$ for real-valued X;
3. ϕ_X uniquely determines the distribution;
4. if $\mathbb{E}[|X|^n] < \infty$ then $\mathbb{E}[X^n] = i^{-n} \phi_X^{(n)}(0)$;
5. $X \perp Y \implies \phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$.

6.2 Common Characteristic Functions

Distribution	$\phi_X(t)$
Bernoulli(p)	$1-p+pe^{it}$
Binomial(n,p)	$(1-p+pe^{it})^n$
Poisson(λ)	$e^{\lambda(e^{it}-1)}$
Normal(μ,σ^2)	$e^{i\mu t-\sigma^2 t^2/2}$
Exponential(λ)	$\lambda/(\lambda-it)$
Cauchy(a,b)	$e^{iat-b\lvert t\rvert}$

The Cauchy row is the point of the whole apparatus: that distribution has no finite moments and therefore no MGF, yet its characteristic function is elementary. Heavy-tailed models — stable laws, variance-gamma, and the jump processes used in option pricing — are almost always specified through ϕ_X for exactly this reason.

6.3 Inversion

For a distribution with density f,

$$f(x)=\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-itx}\phi_X(t)\,dt,$$

so the density can be recovered from the characteristic function. Numerically this is an FFT, which is how Fourier-transform option pricing (Carr–Madan and its descendants) actually computes prices.

6.4 The Central Limit Theorem

Let $X_1,X_2,\dots$ be i.i.d. with $\mathbb{E}[X_i]=\mu$ and $\mathrm{Var}(X_i)=\sigma^2<\infty$, and set $Z_n=\sqrt{n}(\bar{X}_n-\mu)/\sigma$. Then $Z_n\xrightarrow{d}N(0,1)$.

Proof sketch. With $Y_i=(X_i-\mu)/\sigma$ standardised, $\phi_{Z_n}(t)=\left[\phi_Y(t/\sqrt{n})\right]^n$. Since $\mathbb{E}[Y]=0$ and $\mathbb{E}[Y^2]=1$, a second-order expansion gives $\phi_Y(s)=1-s^2/2+o(s^2)$, so

$$\phi_{Z_n}(t)=\left[1-\frac{t^2}{2n}+o(n^{-1})\right]^n\longrightarrow e^{-t^2/2},$$

the characteristic function of $N(0,1)$; Lévy's continuity theorem converts convergence of characteristic functions into convergence in distribution. □

The finite-variance hypothesis is not decorative. When $\sigma^2=\infty$ — the regime suggested by tail-index estimates below 2 — the sum converges instead to a non-Gaussian stable law, and the $\sqrt{n}$ scaling is wrong.

7 Problems

1. Let $X\sim\text{Poisson}(\lambda_1)$ and $Y\sim\text{Poisson}(\lambda_2)$ be independent. Use MGFs to identify the distribution of $X+Y$.
2. Line A runs 70% of the time producing items of mean weight 50 g and variance $4\,\text{g}^2$; line B runs 30% of the time with mean 52 g and variance $9\,\text{g}^2$. Find the overall mean and variance.
3. 5% of a population has a genetic condition. A screen has 90% sensitivity and 95% specificity. Given a positive result, what is the probability the condition is present?
4. Show that if $\phi_X(t)=e^{-|t|}$ then X is standard Cauchy. What characteristic function corresponds to the Laplace density $f(x)=\frac{1}{2}e^{-|x|}$?
5. Let $X\mid\Lambda\sim\text{Poisson}(\Lambda)$ with $\Lambda\sim\text{Gamma}(2,1)$. Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.